AMITY UNIVERSITY KOLKATA

Weekly Progress Report

For the Week Commencing 02 November 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: ETA model training. Trained Random Forest with 300 trees, HistGradientBoosting with depth 8, learning rate 0.05, and 500 iterations.

Achievements for the week

Compared XGBoost with 200 estimators, depth 8, learning rate 0.04, 0.85 row and column subsampling, using GroupShuffleSplit with a 25% test set.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	02 Nov: Prepared the eight ETA feature columns. The training table uses route type, start minute, weekday, progress, off-route distance, speed, profile speed, and support.
Tuesday	03 Nov: Trained the Random Forest baseline. The group split keeps rows from the same session out of both partitions.
Wednesday	04 Nov: Trained HistGradientBoosting and XGBoost. The three regressors were trained against the remaining-time target.
Thursday	05 Nov: Measured MAE, RMSE, R2, and weighted metrics. The metrics expose both ordinary and sample-weighted model behavior.
Friday	06 Nov: Saved model comparisons and learning diagnostics. The saved diagnostics make the model comparison reproducible.